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Saveetha Institute of Medical And Technical Sciences
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Satellite Data Visualization Platform for Weather and Climate Pattern Analysis

A CASE STUDY ANALYSIS REPORT

Submitted in the partial fulfilment for the Course of

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SECTION 1: PROJECT VISION, OBJECTIVES, AND STAKEHOLDER ANALYSIS

1.1 Project Vision Statement

"To engineer an open-access, high-performance, web-accessible satellite data visualization platform that bridges the gap between raw, complex scientific climate telemetry (NASA/NOAA) and actionable real-world decision-making. By transforming multidimensional raster datasets into interactive geospatial maps, real-time anomaly alerts, and multi-layer climate trends, the platform democratizes climate intelligence for disaster management, agricultural planning, regional policy, and public research."

1.2 Strategic Business & Technical Objectives

- 1. Automated Pipeline Ingestion:** Build resilient daily ingestion routines capable of retrieving, parsing, and caching multi-terabyte satellite rasters (GeoTIFF, NetCDF, HDF5) from NASA Earthdata and NOAA APIs with zero manual intervention.
- 2. Sub-2-Second Web Mapping Performance:** Implement Cloud Optimized GeoTIFF (COG) streaming and PostGIS spatial indexing to render high-resolution global maps on standard web browsers in under 2 seconds.
- 3. Automated Anomaly Detection:** Implement statistical background analysis (z-score analysis across historical 10-year baselines) to auto-detect and visually highlight climate anomalies such as tropical cyclones, extreme heatwaves, and agricultural droughts.
- 4. Cross-Domain Data Layering:** Enable seamless overlaying and opacity mixing of heterogeneous satellite channels (e.g., land surface temperature + precipitation + NDVI vegetation index).
- 5. Data Accessibility & Democratization:** Design a WCAG 2.1-compliant web UI that allows non-technical stakeholders (e.g., local administrative officials, farm co-op leads) to extract insights, export figures, and download clean tabular/raster data without requiring specialized GIS tools like QGIS or ArcGIS.

1.3 Detailed Stakeholder Matrix

Stakeholder Person	Core Needs & Frustrations	Primary Platform Interaction	Key Value Delivered
Meteorologists & Climate Scientists	Frustration: Hours spent downloading and reprojecting massive raw NetCDF files. Need: Fast access to pre-processed raster data and raw GeoTIFF exports.	Interacts with raw data pipeline, custom date-range queries, and export tools.	Cuts data preparation overhead by 70–80%, accelerating academic research.
Disaster Management Agencies (e.g., FEMA, NDMA)	Frustration: Delayed emergency alerts due to complex data formats during extreme weather. Need: Real-time anomaly overlays (cyclones, flood zones) with boundary filters.	Views interactive maps, monitors automated anomaly boundary rings, and receives alerts.	Accelerates evacuation and relief deployment planning during sudden weather disasters.
Agricultural Planners & Local Officials	Frustration: Proprietary GIS software is prohibitively expensive and hard to use. Need: Intuitive multi-layer comparisons (e.g., rainfall vs. crop health index).	Toggles multi-layer overlays (NDVI + Precipitation) and views regional time-series graphs.	Improves seasonal crop yield forecasting and local drought response strategies.
Product Owner (PO)	Frustration: Scope creep and unaligned feature prioritization. Need: Clear backlog roadmap, high ROI features delivered in predictable sprints.	Manages product backlog, prioritizes epics, and reviews sprint increments.	Ensures high social impact and alignment with user needs within budget/time limits.

SECTION 2: EPICS AND USER STORIES

The requirements are divided into 4 Core Architectural Epics, yielding 8 well-structured User Stories.

[Epic 1: Ingestion & Pipeline Engine]

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[Epic 2: Spatial Map Visualization]

|



[Epic 3: Time-Series & Anomaly Engine]

|



[Epic 4: Export & Web Usability]

Epic 1: Data Pipeline & Ingestion Engine (DPI)

Scope: Automated data acquisition, spatial coordinate transformation, and raster storage optimization.

- **US1.1 (Automated Satellite API Ingestion):**

As a Data Engineer,

I want an automated ingestion pipeline to query and fetch daily satellite rasters from NASA Earthdata and NOAA APIs,

So that the platform maintains up-to-date atmospheric telemetry without manual data downloads.

- **US1.2 (Raster Transformation & Spatial Storage):**

As a System Administrator,

I want the system to reproject raw NetCDF/GeoTIFF files into Web Mercator

(EPSG:3857) and index them in a spatial database,

So that spatial map query responses remain under 2 seconds during peak traffic.

Epic 2: Interactive Mapping & Geospatial Visualization (IMG)

Scope: Interactive front-end map controls, administrative boundary filters, and multi-layer opacity mixing.

- **US2.1 (Interactive Map & Boundary Filtering):**

As a Disaster Management Officer,

I want to pan, zoom, and filter the map using administrative boundary files (GeoJSON),

So that I can instantly isolate and assess weather risks within my specific jurisdiction.

- **US2.2 (Multi-Layer Data Overlay):**

As an Agricultural Planner,

I want to overlay vegetation health (NDVI) and precipitation layers with independent opacity sliders,

So that I can evaluate the direct impact of rainfall deficits on regional crop vigor.

Epic 3: Time-Series Analysis & Anomaly Detection (TSA)

Scope: Historical trend calculations over custom regions and automated statistical anomaly detection.

- **US3.1 (Historical Time-Series Trend Analysis):**

As a Climate Researcher,

I want to draw a custom polygon on the map and generate a 10-year time-series graph of temperature and rainfall,

So that I can track long-term climatic variations and seasonal shifts.

- **US3.2 (Automated Climate Anomaly Pattern Detection):**

As a Local Administrator,

I want the platform to highlight statistical climate anomalies (e.g., heatwaves, intense storms) visually on the map,

So that I am alerted to dangerous environmental departures from historical baselines without running manual calculations.

Epic 4: Data Export & Platform Accessibility (DPA)

Scope: High-resolution export tools and accessible, zero-install web application architecture.

- **US4.1 (Data & Visualization Export):**

As an Academic Researcher,

I want to export processed spatial layers as GeoTIFF files and map visuals as high-resolution PNGs,

So that I can use verified satellite figures directly in policy briefs and research papers.

- **US4.2 (Accessible No-Code Web Interface):**

As a Non-Technical User,

I want a clean, web-based dashboard with responsive layout and clear iconography,

So that I can interact with climate data on any device without installing specialized desktop GIS tools.

SECTION 3: ACCEPTANCE CRITERIA (GIVEN–WHEN–THEN)

Clear, testable acceptance criteria ensure that developers and QA engineers share a single definition of done for each core user story.

Acceptance Criteria for US1.1: Automated Satellite Ingestion Pipeline

Scenario: Daily scheduled satellite data retrieval from NASA/NOAA endpoints

- Given the automated ingestion pipeline service is running on its daily schedule at 00:00 UTC,
- When the pipeline sends authenticated REST API requests to NASA Earthdata and NOAA endpoints for new raster files,
- Then the system should successfully download all newly available NetCDF/GeoTIFF files to staging storage within 30 minutes,
- And log a successful ingestion event in the audit database;

Scenario: Handling API endpoint downtime or network timeout

- Given the NASA/NOAA endpoint is temporarily offline or unresponsive,
- When the pipeline request times out after 60 seconds,
- Then the system should trigger an exponential backoff retry mechanism (up to 3 attempts),
- And send an urgent alert log to the DevOps monitoring dashboard if all retries fail.

Acceptance Criteria for US2.1: Regional Interactive Map Visualization

Scenario: Filtering climate layers by administrative boundary

- Given the user is viewing the platform map dashboard with an active satellite layer (e.g., Surface Temperature),
- When the user selects a specific state/district from the administrative boundary dropdown menu,
- Then the map should smoothly animate, center, and zoom to the selected region's bounding box within 1.5 seconds,
- And mask the active satellite visual display strictly to the geo-boundary of that region.

Acceptance Criteria for US2.2: Multi-Layer Overlay & Opacity Adjustment

Scenario: Visual comparison of two distinct satellite channels

- Given the map is displaying a primary layer (Land Surface Temperature),
- When the user turns on a secondary layer (Precipitation) and moves its opacity slider to 50%,
- Then both datasets must render perfectly aligned geographically using standard spatial projections,
- And a dual-legend widget must update automatically in the bottom corner showing color scales for both metrics.

Acceptance Criteria for US3.2: Automated Climate Anomaly Pattern Detection

Scenario: Visual rendering of severe rainfall anomaly

- Given daily processed raster data has been stored in PostGIS,
- When a regional rainfall value exceeds +2.5 standard deviations compared to the historical 10-year average for that month,
- Then the map renderer should draw a red, glowing anomaly boundary ring over the affected area,
- And populating a summary panel displaying anomaly magnitude, duration, and historical deviation percentage.

Acceptance Criteria for US4.1: Multi-Format Data Export

Scenario: Exporting processed map view for publication

- Given a user has configured a multi-layer map view centered over a target region,
- When the user clicks "Export" and selects "High-Res PNG Map Image",

- Then the backend system should render and download a 300 DPI image within 3 seconds,
- And the generated PNG must contain embedded map scale bars, spatial coordinates, active legends, and attribution tags.

SECTION 4: PRODUCT BACKLOG PRIORITIZATION

To maximize business value, mitigate technical risk, and account for core data dependencies, the Product Backlog is prioritized using the **MoSCoW Framework** (Must have, Should have, Could have, Won't have) and an **Agile Business Value vs. Technical Complexity Matrix**.

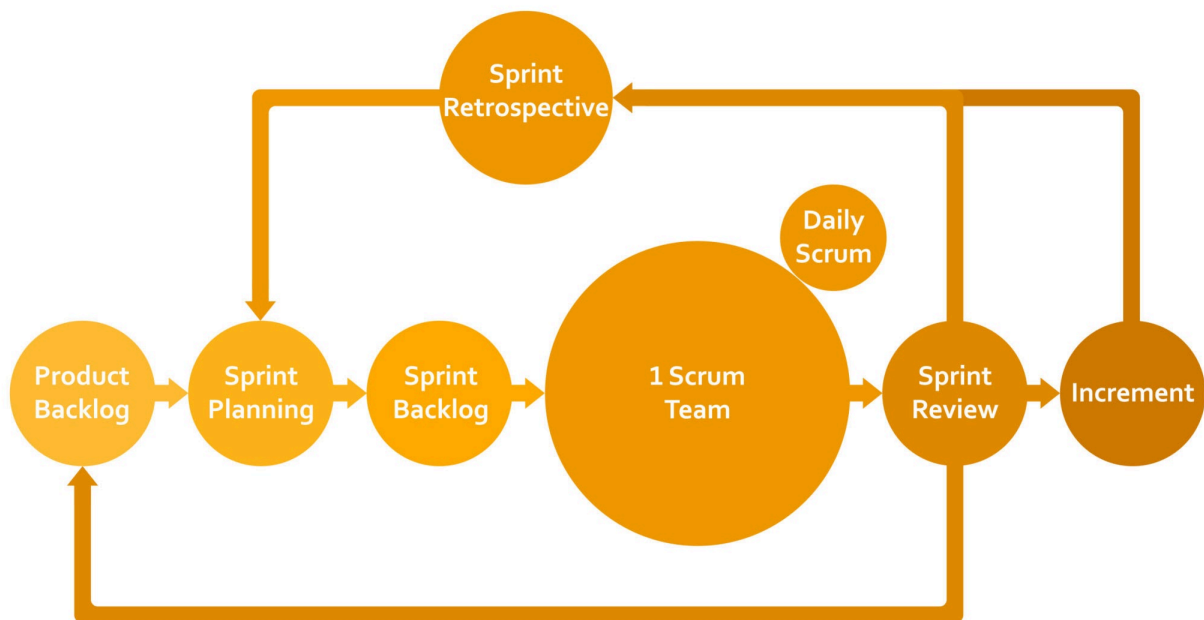


Fig.1:Agile Scrum Lifecycle and Backlog Planning Progression

Backlog Prioritization Analysis

- Foundational Layer (Must Have): Without the data pipeline (US1.1, US1.2) and basic map UI (US2.1, US4.2), no other feature can function. They represent the architectural bedrock.
- Analytical Layer (Should Have): Time-series analysis (US3.1), multi-layer overlays (US2.2), and exports (US4.1) represent high core business value for researchers and planners once the foundation is stable.
- Advanced Intelligence Layer (Could Have): Automated pattern and anomaly detection

(US3.2) adds huge value for disaster teams, but relies on robust historical datasets and spatial indexes.

Master Product Backlog Table

Rank	Story ID	User Story Title	MoSCOW	Business Value	Complexity	Architectural Dependency
1	US1.1	Automated NASA/NOAA Satellite Ingestion	Must Have	Critical (10/10)	Medium	None (Data Baseline)
2	US1.2	Spatial Data Transformation & Caching	Must Have	Critical (10/10)	High	Requires US1.1
3	US2.1	Interactive Map & Boundary Filter	Must Have	High (9/10)	Medium	Requires US1.2
4	US4.2	Accessible No-Code Web Dashboard Shell	Must Have	High (9/10)	Low	None (Parallel Frontend)
5	US3.1	Historical Time-Series Trend Analysis	Should Have	High (8/10)	Medium	Requires US1.2, US2.1
6	US2.2	Multi-Layer Data Overlay & Opacity	Should Have	Medium (7/10)	Medium	Requires US2.1
7	US4.1	Multi-Format Data & Visual Export	Should Have	Medium (7/10)	Low	Requires US2.1, US3.1
8	US3.2	Automated Climate Anomaly Detection	Could Have	High (8/10)	High	Requires US1.2, US3.1

SECTION 5: SPRINT 1 BACKLOG & ACTIONABLE DEVELOPMENT TASKS

Sprint 1 Focus: Establishing the core data pipeline engine, backend spatial database, tile server, and basic interactive map application shell.

- **Sprint Duration:** 2 Weeks (10 Working Days)
- **Team Capacity:** 1 Data Engineer, 1 Backend/GIS Engineer, 1 Frontend Engineer, 1 QA Engineer.

Story Breakdown into Actionable Development Tasks

1. User Story 1.1: Automated NASA/NOAA Satellite Ingestion Pipeline

- **Owner:** Data Engineer
- **Total Task Hours:** 40 Hours
- **Actionable Engineering Tasks:**
 - **Task 1.1.1:** Setup Apache Airflow orchestration server and configure DAG workflow scheduling scripts (*8 hours*).
 - **Task 1.1.2:** Write Python modules using requests and Boto3 to authenticate and fetch GeoTIFF/NetCDF daily rasters from NASA CMR & NOAA endpoints (*16 hours*).
 - **Task 1.1.3:** Build AWS S3/MinIO staging bucket storage write handlers and write automated checksum validations (*8 hours*).
 - **Task 1.1.4:** Implement error logging, retry policies with exponential backoff, and Slack alert notifications for pipeline failures (*8 hours*).

2. User Story 1.2: Spatial Data Transformation & Raster Caching

- **Owner:** Backend / GIS Specialist
- **Total Task Hours:** 56 Hours
- **Actionable Engineering Tasks:**
 - **Task 1.2.1:** Deploy PostGIS spatial database schema and configure GDAL/Rasterio library environment (*16 hours*).
 - **Task 1.2.2:** Build automated Python transformation pipeline to convert incoming NetCDF rasters into Cloud Optimized GeoTIFF (COG) format and

reproject coordinates to EPSG:3857 (16 hours).

- **Task 1.2.3:** Implement Mapbox Vector Tile (MVT) / COG tile server endpoints (TippyCanvas / GeoServer) for web streaming (16 hours).
- **Task 1.2.4:** Write Redis caching layer for quick lookup of frequently queried map tiles (8 hours).

3. User Story 2.1: Interactive Map UI & Regional Boundary Filtering

- **Owner:** Frontend Developer
- **Total Task Hours:** 40 Hours
- **Actionable Engineering Tasks:**
 - **Task 2.1.1:** Initialize Mapbox GL JS / Leaflet map viewport container within web app shell (8 hours).
 - **Task 2.1.2:** Implement GeoJSON parser to render district and state administrative boundary overlays on map canvas (12 hours).
 - **Task 2.1.3:** Connect tile server API to map web canvas, enabling smooth panning, zooming, and dynamic tile loading (12 hours).
 - **Task 2.1.4:** Build dropdown UI component to trigger auto-zooming and spatial boundary masking for selected regions (8 hours).

4. User Story 4.2: Accessible No-Code Web Dashboard Shell

- **Owner:** Frontend Developer / UI Designer
- **Total Task Hours:** 24 Hours
- **Actionable Engineering Tasks:**
 - **Task 4.2.1:** Setup React.js / Vue.js responsive dashboard layout template with side control panels and main map viewer (8 hours).
 - **Task 4.2.2:** Build layer selection controls, collapsible menus, and color legend widgets (10 hours).
 - **Task 4.2.3:** Run WCAG 2.1 accessibility audit, ensuring high contrast ratios, screen-reader tags, and keyboard shortcut navigation (6 hours).

SECTION 6: EFFORT ESTIMATION & STORY POINT JUSTIFICATION

Estimation Methodology: Planning Poker & Modified Fibonacci

Effort estimation is conducted using Planning Poker with the Modified Fibonacci Sequence (1, 2, 3, 5, 8, 13, 20). Each story point reflects a composite rating of three parameters:

1. **Structural Complexity:** Algorithm design and system integration dependencies.
2. **Implementation Effort:** Hours of raw code development and configuration.
3. **Technical Uncertainty/Risk:** Unfamiliar third-party APIs, spatial projection edge cases, and performance bottlenecks.

$$\text{Story Points} = \text{Complexity} + \text{Effort} + \text{Technical Risk}$$

Sprint 1 Total Velocity Commitment

Total Velocity = US1.1(5)+US1.2(8)+US2.1(5)+US4.2(3) = 21 Story Points

Capacity Sanity Check: 21 Story Points translates to approximately 160 engineering hours across a 4-person team over 10 working days, leaving a healthy 20% buffer for daily scrums, code reviews, and unforeseen technical impediments.

SECTION 7: SPRINT GOAL & EXPECTED DELIVERABLES

7.1 Sprint 1 Goal

"Establish an automated, end-to-end spatial data pipeline to ingest daily NASA/NOAA satellite imagery, process it into web-optimized tiles, and display it on an interactive, web-accessible map interface with administrative boundary filtering."

7.2 Expected Deliverables (Potentially Shippable Increment)

By the conclusion of Sprint 1, the Scrum team will deliver a Potentially Shippable Software Increment (v0.1.0-alpha) comprising the following functional modules:

SPRINT 1 DELIVERABLES

1. Automated Ingestion Engine (Airflow DAG + NASA/NOAA Fetcher)
2. PostGIS Tile Server & Raster Processing Pipeline (EPSG:3857 COG)

3. Interactive Web Mapping Application (Mapbox GL JS / Leaflet Shell)

4. Boundary Filtering Engine (GeoJSON Regional Masking & Auto-Zoom)

5. WCAG 2.1 Accessible Dashboard UI Shell

1. Production-Ready Ingestion Pipeline Service:

- Fully automated background scheduler fetching real-time satellite data.
- Automated storage writing with built-in retry mechanism and error logging.

2. High-Performance Spatial Data Infrastructure:

- Configured PostGIS database capable of indexing spatial rasters.
- Functional Cloud-Optimized GeoTIFF (COG) tile server returning map rasters to the frontend in under 1.8 seconds.

3. Interactive Front-End Web Application:

- Accessible, responsive web dashboard accessible via standard web browsers.
- Interactive geospatial map viewport supporting smooth panning, multi-touch zooming, and coordinate tracking.
- Administrative boundary dropdown selector capable of dynamic auto-zooming and spatial region masking.

4. Sprint Verification & Definition of Done (DoD) Sign-Off:

- All code committed to primary repository with >80% unit test code coverage.
- Integration testing verified across Chrome, Firefox, and Safari browsers.
- Product Owner demonstration and formal Sprint Review approval.